Jay Bhatt

Edmonton, AB, Canada

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SKILLS

Languages: Python, C, C#, Rust, Kotlin, Go, Odin, Bash
Technologies: Linux/Unix, Git, Unraid, Proxmox, Docker, SQL (Microsoft, MySQL, SQLite), Jira

RESEARCH

Research Assistant *Edmonton, AB*
Grant MacEwan University *Sept 2026 - Present*

- Continuing development of the normative modeling pipeline for structural MRI in ALS patients from the CALSNIC dataset

Undergraduate Research Assistant *Edmonton, AB*
Grant MacEwan University *May 2026 - Aug 2026*

- Building a normative modeling pipeline to analyze structural MRI in ALS patients from the CALSNIC dataset
- Applied pretrained lifespan Bayesian linear regression models to FreeSurfer cortical thickness and subcortical volume features, learning site specific offsets via adaptation to compute per region Z scores and centiles
- Quantified patient-vs-control deviations with Cohen's d and Welch t-tests under Benjamini-Hochberg FDR correction, and validated model calibration on held-out healthy controls
- Awarded an SRS grant to fund this research
- Won first place in the SRS presentation competition

EXPERIENCE

Pharmacy Assistant *Edmonton, AB*
Ellerslie Pharmacy *March 2025 - Present*

- Ensured strict regulatory compliance in prescription processing, packaging, and labeling; direct experience with data accuracy and governance standards
- Maintained patient records in pharmacy management software while upholding data integrity and privacy requirements
- Identified and resolved technical risks in service computers, reducing potential data loss and operational downtime

Lunchtime Supervisor *Edmonton, AB*
Jan Reimer School *April 2022 - Present*

- Maintained safe and documented environments; escalated incidents through appropriate reporting chains

PROJECTS

Ancient DNA Classifier *Private Repo*

- Built and evaluated traditional machine learning and deep learning models to classify human mitochondrial DNA as ancient or modern
- Implemented a suite of ML algorithms, including Logistic Regression, Random Forest, Gaussian Naive Bayes, SVM, KNN, XGBoost, and CatBoost.
- Designed and trained a deep learning model using PyTorch

F1 Lap Time Prediction *Private Repo*

- Developed an end-to-end machine learning pipeline to predict Formula 1 lap times, managing feature engineering and model evaluation across the full ML lifecycle.

EmBlocks *github.com/IsaacMcCracken/EmBlocks-HackED-2026*

- Developed a block-based IDE and C-level abstraction layer for the RP2040 microcontroller, enabling simplified embedded system development through Scratch-like visual programming.

Fluid Simulation *github.com/jay-a-bhatt/Fluid-Simulation-hackED-2025*

- Developed a high-performance fluid simulation using Rust and WebGPU, implementing PIC/FLIP algorithms for realistic, real-time particle dynamics.

- Optimized for cross-platform web deployment using WebAssembly (WASM), featuring a responsive UI with custom controls for interactive user manipulation.

Home Server

- Self-host and maintain a 15+ service environment (Unraid, Proxmox, Docker), managing secure access and control
- Hardened infrastructure using Private Key SSH, CrowdSec IPS, and MFA-backed SSO (Authentik) to enforce strict identity-aware access control.

EDUCATION

Grant MacEwan University, Edmonton, AB *2026 Expected*

BSc Computer Science Major + Biology Minor

- **CMPT Courses:** Software Engineering, Machine Learning, Robotics, Databases, Networking, Cryptography, Operating Systems, Human/Computer Interaction, Data Structures & Algorithms, Numerical Methods
- **BIOL Courses:** Eukaryotic Cell Biology, Genetics, Evolution, Ecology

W.P. Wagner High School, Edmonton, AB *June 2021*

High School Diploma